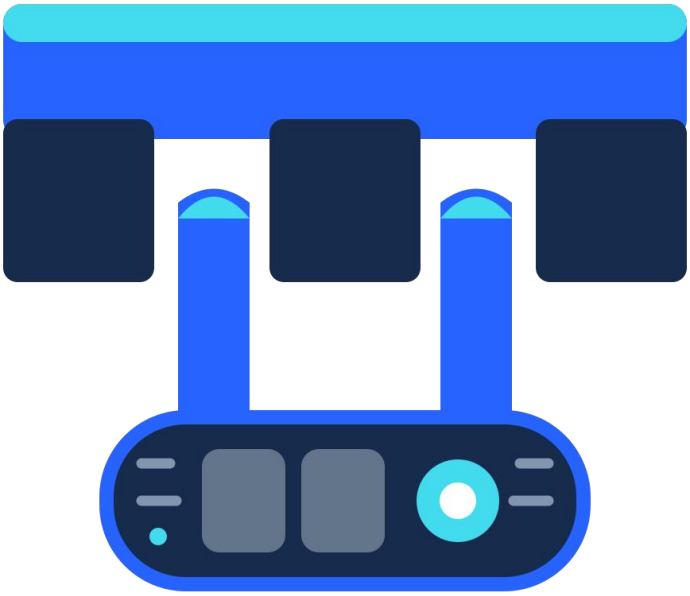
MRDVS® 迈尔微视



AW3 Smart Docking

Pallet Docking Solution White Paper

Standard international offering: pure 3D geometric positioning
2D semantic positioning is not included

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V0.2 · 2026-11-12

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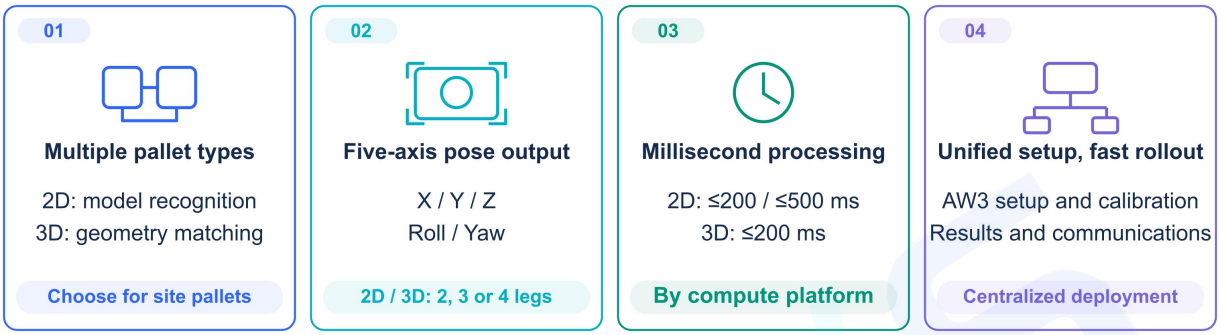
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1 Product Overview

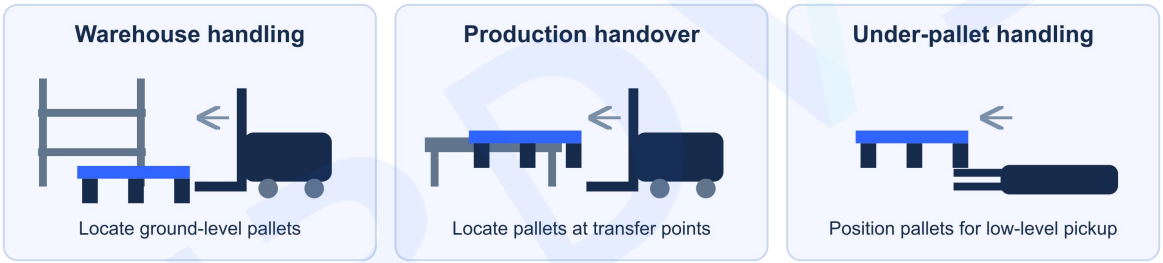
Smart Docking runs on the AW3 Algorithm Platform. It identifies the pallet fork-entry face and supplies the reference point position, orientation and positioning status to autonomous forklifts.

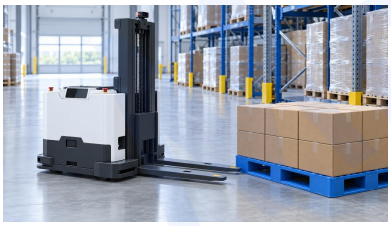
AW3 manages devices, configuration and communication. Pallet Docking provides perception; the vehicle selects targets and controls motion and safety.

1.1 Key Advantages



1.2 Application Scenarios





Warehouse pallet handling

Locate pallets in floor-level buffer areas and provide the position and angle for autonomous fork entry.



Production-line transfer

Locate pallets at production transfer points for handover between equipment and handling vehicles.



Under-pallet fork handling

Locate the low-level fork-entry face for under-pallet fork robots.

Solution and Computing Options

The standard international offering uses pure 3D geometric positioning. It does not include 2D semantic positioning.

For both S11 2D and pure 3D, use the customer's existing AGV industrial PC or computing platform first. No dedicated board is needed if it meets the requirements of the selected mode.

S11 · 3D | Customer AGV compute (international recommendation)

Pure 3D geometry · Runs on the customer's Linux industrial PC

Suitable PC: no extra board. Optional board after assessment and agreement, at extra cost.

M4 Mega | Onboard compute

Pure 3D geometry · Runs inside the camera

No external computing device required

S11 · 2D | Customer AGV compute (not standard internationally)

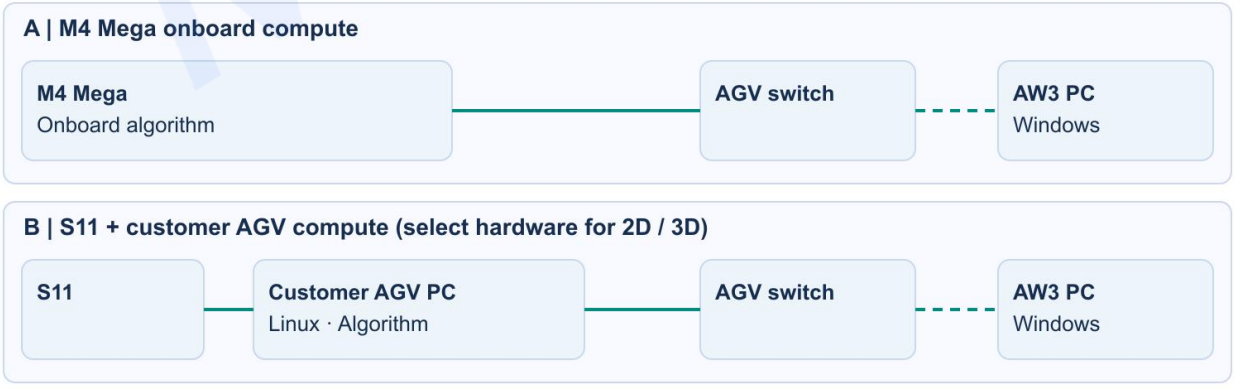
Customer platform must use RK3576 / RK3588; site RGB training data required

Insufficient compute: optional board after assessment and agreement, at extra cost.

S11 compute: 2D requires RK3576/RK3588; pure 3D recommends Intel Core i5-class or RK3568 and above. Linux and algorithm compatibility must be confirmed; lower 3D specifications require assessment. CPU/GPU usage affects processing time. For either mode, insufficient capacity may be addressed with an additional compute board, subject to assessment and agreement, at extra cost.

AW3 PC: Windows, for configuration, monitoring and troubleshooting. Connect when needed; it need not remain on the vehicle. The AGV industrial PC running the algorithm must remain powered and connected.

M4 Mega layout: pure 3D. Select S11 compute for 2D or 3D. Solid links stay connected; the dashed AW3 PC link is used as needed.



1.3 Cameras and Computing Platforms

S11 supports both low-profile under-pallet forklifts and forklift AGVs. Its small blind zone and wide field of view suit low mounting positions and close-range pallet recognition.

S11 | Low mounting, close range

Small blind zone: depth sensing from 0.1 m
Wide depth FOV: 136° × 56°
Depth accuracy: ≤1 cm
Wide reflectivity range: 10%–90%
Ambient light immunity: 100 kLux

M4 Mega | Range and onboard compute

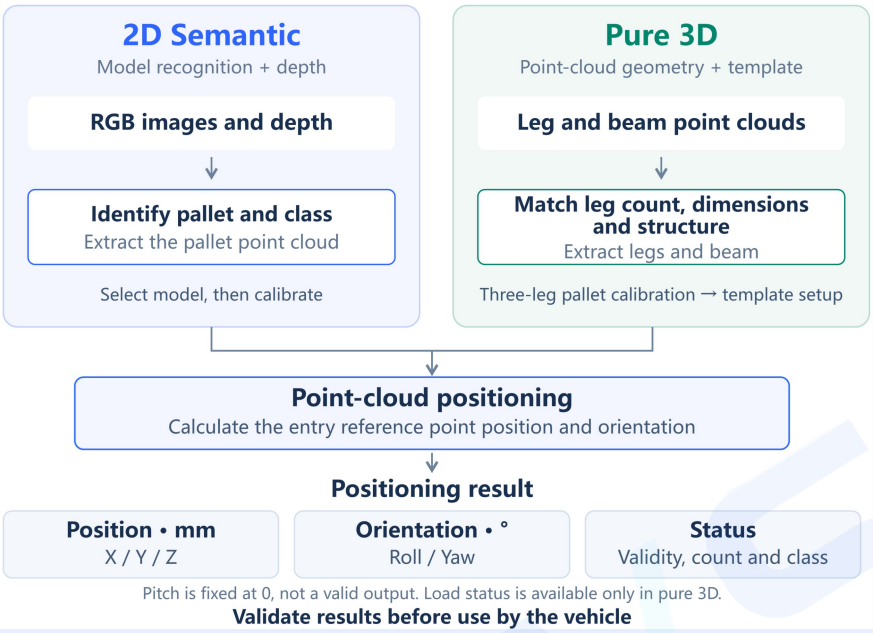
Pallet recognition range: up to 3.0 m
Detailed depth data: 640 × 480 pixels
1 TOPS onboard compute; no external board
Global-shutter RGB with RGB-D alignment
IP67; Gigabit Ethernet / CAN / I/O

Parameter	S11	M4 Mega
Computing platform	Customer AGV compute Recommended for 2D / 3D	Onboard compute, 1 TOPS
Depth technology / illumination	dToF / 940 nm	iToF / 940 nm
Output Data	Depth, RGB, IR amplitude	Depth, RGB, amplitude
Depth Resolution / Frame Rate	240 × 96 Typical 10 / max. 15 fps	640 × 480 Typical 15 / max. 25 fps
RGB Resolution / Frame Rate	1280 × 1080 Typical 10 / max. 15 fps	1280 × 960 Typical 15 / max. 25 fps
TOF Field of View (H×V)	136° × 56° (±3°)	81° × 61° (±3°)
RGB Field of View (H×V)	120° × 110° (±3°)	86° × 55° (±3°)
Depth Measurement Range[1]	Outdoor: 0.1–5 m Indoor: up to 10 m	0.2–5 m
Measurement Accuracy[2]	≤ 1 cm	±3 mm + 0.25% × depth
Power Supply	Ethernet: 12–28 VDC USB: 5 V / USB 3.0	24 VDC / 2 A
Average Power Consumption[3]	< 4 W	6.8 W @ 24 V
Communication Interfaces	Ethernet / USB	Gigabit Ethernet / CAN / 4 IO
Dimensions	90 × 25 × 25 mm	92 × 47 × 51 mm
IP Rating	Ethernet version: IP54	IP67
Operating Temperature	−20–60°C	−20–60°C

[1] Specification conditions: target reflectivity 10–90%. [2] Camera measurement accuracy is not pallet positioning accuracy. [3] S11 power excludes external computing hardware.

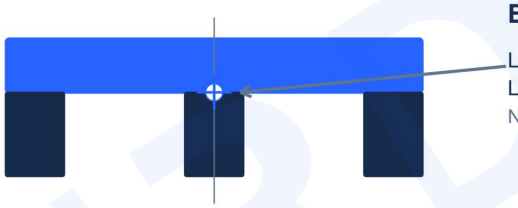
2 How It Works

2.1 Two Positioning Methods



The standard international offering uses pure 3D geometric positioning. It does not include 2D semantic positioning.

Entry Face Front View



Entry Reference Point

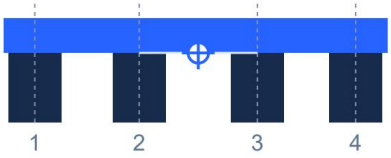
Lateral center of entry face
Lower edge of the upper beam
Not the volume or detection-box center

Two-leg output reference point



Midpoint between the two leg centers

Four-leg output reference point



Sort legs laterally; use the midpoint between the two inner leg centers

Benefits and setup trade-offs

2D Semantic Positioning

Benefits

Train models for different site pallet types.
Classify pallets with or without a lower beam.

Preparation and maintenance

Collect site-pallet RGB data for semantic training.
Validate new types; retrain if needed.

Pure 3D Positioning

Benefits

No model training required.
Similar pallet geometry can share a template.

Preparation and maintenance

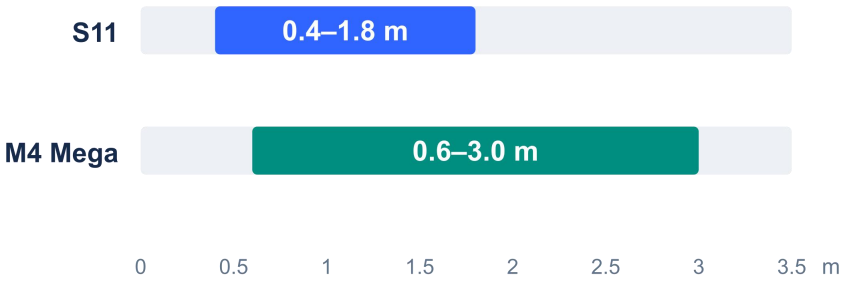
Tune parameters to site-pallet dimensions.
Many distinct types may need frequent template changes or parameter adjustments.

3D is an alternative method, not an automatic failover when 2D fails. Calibrate, configure and validate it before use.

3 Performance and Operating Range

3.1 Detection Distance and Processing Time

Recognition Distance Range



Select the range for the camera. The usable range varies with pallet size, type, material and model; validate it with site pallets.

Algorithm processing time

2D model-based recognition		3D geometry matching	
RK3576 / RK3588	**200 ms max.**	S11 + external compute	**200 ms max.**
M4 Mega onboard	**500 ms max.**	M4 Mega onboard	**200 ms max.**

Algorithm time excludes acquisition, communication and vehicle control. External-platform times depend on CPU/GPU usage. Validate the complete cycle on site.

3.2 Recognition and Outputs

2D Semantic Positioning

- **2-, 3- and 4-leg pallets**
  With a suitable model
- **With / without lower beam**
- **Multiple pallets**

No load detection

Pure 3D Positioning

- **2-, 3- and 4-leg pallets**
  Geometry templates: presets / custom
  Add, edit, delete or switch templates
- **Load detection when enabled**
- **Stacked pallets**

No lower-beam classification

Shared outputs: five-axis pose and positioning status
X / Y / Z (mm) · Roll / Yaw (°); Pitch is fixed at 0
Use current valid results only. The vehicle selects the target.

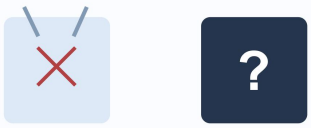
Positioning and docking metrics

X / Y position error	Z within calibrated range	Yaw / Roll angle error	Horizontal yaw range
≤15 mm	**≤10 mm**	**≤2°**	**±15°**

Vision positioning metrics; final vehicle docking accuracy also depends on motion control and mechanical tolerances.

3.3 Operating Conditions and Limits

Pallet material



Reflective surfaces are unsuitable.
Assess black or low-reflectance pallets.

Pallet structure and visibility



Show the full entry face, legs and beams.
Occlusion and missing parts affect positioning.
Thin-legged, deformed or self-built pallets require assessment before use.

2D model suitability



Train with RGB images of site pallets.
Coverage depends on training and validation.

Entry-face angle



Within 15° relative to the camera.
Angles over 15° are filtered.

3D recognition height



Camera-to-floor height must match calibration.
A height change requires ground recalibration.

3D roll tilt

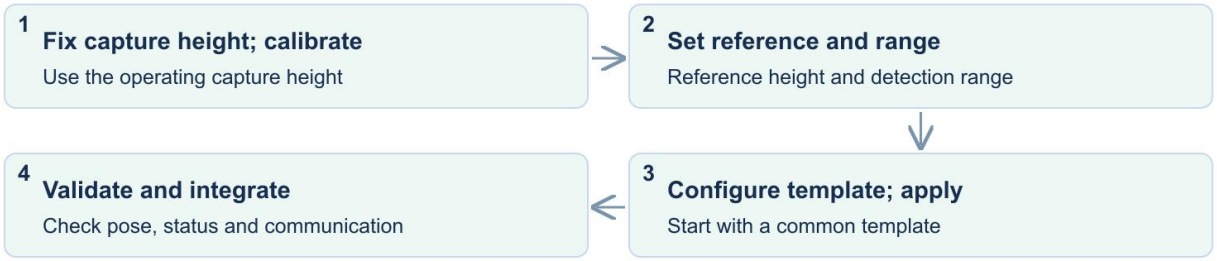


Roll tilt may prevent recognition.
Roll output does not mean all tilts are supported.

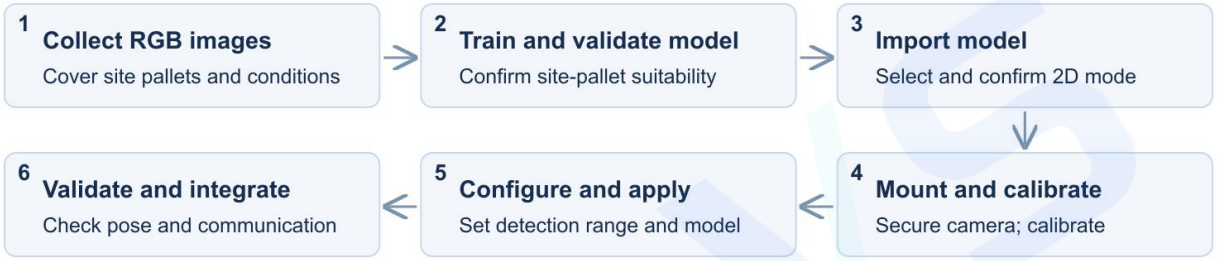
4 Deployment and Configuration

4.1 Deployment Workflow

3D | Standard international offering



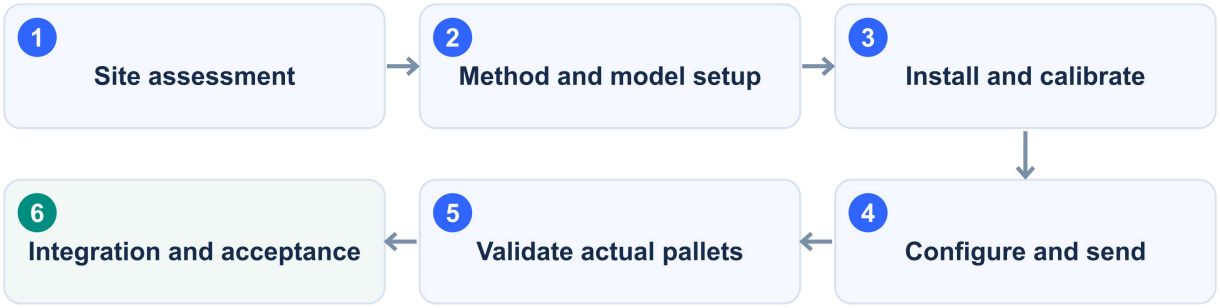
2D | Not included in the standard international offering



International projects use the pure 3D workflow above. See “Solution and Computing Options” for computing requirements and optional-board charges. CPU/GPU usage affects processing time; validate the complete cycle on site.

5 Validation and Acceptance

Agree on project criteria for positioning error, repeatability, recognition performance and detection cycle time before acceptance. Verify them using the workflow and checks below.



Acceptance item	Coverage
Recognition	Actual pallet types/materials, near/far distances, angles and occlusion; check missed and false detections.
Positioning error	Compare X, Y, Z, Roll and Yaw against references; check repeatability.
Response time	Validate algorithm time and the complete system cycle separately. Include acquisition, communication and vehicle control; set waits/timeouts under actual resource load, with margin.
Functions / communication	Validate multiple-pallet detection in the mode used. Test load status only in 3D with load detection enabled. Check target identity, status, units and result updates at the receiver.
Recovery	Test no pallet, invalid results, disconnection and recovery; reject invalid or stale data.

6 Frequently Asked Questions

Question	Answer
Does 2D mean planar output only?	No. “2D” refers to image-model recognition. Depth point clouds still provide 3D position and usable orientation.
Can one 2D model cover all pallets?	No. Train on site pallets and validate actual materials, distances, angles and lighting.
Does 3D automatically replace failed 2D?	No. Select and configure 3D separately, then calibrate and validate it before use.
Which camera reaches farther?	M4 Mega: 0.6–3.0 m. S11: 0.4–1.8 m. S11 requires external computing; see Chapter 4. Validate the usable range with site pallets and models.
Are black and reflective pallets supported?	Reflective surfaces are unsuitable. Evaluate black pallets with the selected camera and actual site samples; appearance alone does not establish suitability.
How are 2-, 3- and 4-leg pallets supported?	Both modes support them. Use a suitable model for 2D or the matching leg-count template for 3D. Pure 3D near/far calibration still requires a three-leg pallet.
What do Closed and Open pallet mean?	Here, Closed has a continuous bottom board joining the legs below the fork openings (With Lower Beam); Open has none (No Lower Beam). This 2D classification applies only to the entry face visible to the camera, not the whole pallet base.
Is real-time motion feedback supported?	No. Algorithm time is not the complete cycle. Include acquisition, communication and vehicle control when setting waits/timeouts from on-site measurements. Reject stale results.
Does load detection measure weight?	No. Only 3D, with load detection enabled, reports Empty or Loaded. It does not replace weighing or safety checks. 2D does not support load detection.

Revision History and Support

Version	Date	Description
V0.1	2026-09-28	Measured performance, camera specifications, applications and operating limits.
V0.2	2026-11-12	Clarified the international pure 3D offering; added S11 industrial-PC configurations, optional-board charges and connections; updated deployment workflows. Clarified customer AGV compute as the first choice for both S11 modes, with optional paid boards if capacity is insufficient.

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